

CALCULUS 1

1. LIMITS AND CONTINUITY

Exercise 1.1. Compute the following limits and indicate which properties you are using.

(a) $\lim_{x \rightarrow 1} (x^2 + 3x + 2)\sqrt{3x - 1}$

(h) $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2}$

(b) $\lim_{x \rightarrow 0} \ln(x + 1)$

(i) $\lim_{h \rightarrow 0} \frac{(t + h)^2 - t^2}{h} \quad t \in \mathbb{R}$

(c) $\lim_{x \rightarrow -1} \frac{e^{2x+5}}{x + 2}$

(j) $\lim_{x \rightarrow 2} \frac{-5x^2 + 9x + 2}{x^2 - 4}$

(d) $\lim_{x \rightarrow 3} \frac{x^2 - 1}{x^2 - 3x + 2}$

(k) $\lim_{x \rightarrow b} \frac{x^3 - b^3}{x - b}$

(e) $\lim_{x \rightarrow -2} \frac{x}{x + 3} - \frac{x^2 + x}{\sqrt{x^2 + 5}}$

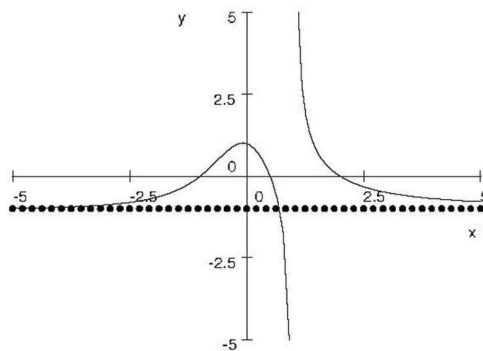
(l) $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$

(f) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{4x+1} - 3}$

(m) $\lim_{x \rightarrow 0} \sqrt{\frac{x+4}{x}} - \frac{2}{\sqrt{x}}$

(g) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin^2 x + \tan x}{\cos(\frac{2x}{3})}$

Exercise 1.2. Consider the function $y = f(x)$ whose graph is



- (1) Find the domain of f and the limits on the endpoints of its domain.
- (2) How many solutions does the equation $f(x) = -1$ have?
- (3) How many solutions does the equation $f(x) = 0$ have?
- (4) How many solutions does the equation $f(x) = m$ have? Consider all the possibilities for a real number m .

Date: January 11, 2021.

Exercise 1.3. Compute the following limits:

$$(a) \lim_{x \rightarrow +\infty} \frac{5x^4 - x^3 + 2}{3x^4 - 3x^2 + 1}$$

$$(b) \lim_{x \rightarrow \infty} \frac{8x^3 + 2x^2}{3x^2 - 2x + 2}$$

$$(c) \lim_{x \rightarrow +\infty} \frac{x^3 + 3x + 1}{2x^4 + 2x^2 + 1}$$

$$(d) \lim_{x \rightarrow +\infty} \frac{3x^2 + 1}{-7x^2 + 3x + \sqrt{x}}$$

$$(e) \lim_{x \rightarrow +\infty} (x^7 - 10x^3 + 3)$$

$$(f) \lim_{x \rightarrow +\infty} (-x^6 + x^5 + \sqrt{x})$$

$$(g) \lim_{x \rightarrow \infty} \frac{x^7 - \sqrt{3}^{10}}{x} + 9x^7$$

$$(h) \lim_{x \rightarrow +\infty} \frac{\sqrt{x^3 + 1} - x}{x + 5}$$

$$(i) \lim_{x \rightarrow +\infty} (\ln x - 2 \ln(x + 1))$$

$$(j) \lim_{x \rightarrow +\infty} \frac{3^{x+1} + 2}{3^x - 5}$$

Exercise 1.4. Compute the following limits:

$$(a) \lim_{x \rightarrow 0} x \sin \frac{1}{x}$$

$$(b) \lim_{x \rightarrow 0} \frac{x^2}{2} \cos(x + \frac{1}{x})$$

$$(c) \lim_{x \rightarrow 2} (x^2 - 4) \sin(\frac{1}{x - 2})$$

$$(d) \lim_{x \rightarrow 0} x \sin(h(x))$$

$$(e) \lim_{x \rightarrow 0} \sin x \cdot \cos[\ln(1 + \frac{1}{x})]$$

$$(f) \lim_{x \rightarrow 0} \sqrt{x} \cdot e^{\cos \frac{1}{x}}$$

Exercise 1.5. Using that $\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$, compute:

$$(a) \lim_{x \rightarrow 0} \frac{x}{\sin x}$$

$$(b) \lim_{x \rightarrow 0} \frac{\tan x}{x}$$

$$(c) \lim_{x \rightarrow 0} \frac{\sin(3x)}{2x}$$

$$(d) \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 2x}$$

$$(e) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

$$(f) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\frac{\pi}{2} - x}$$

$$(g) \lim_{x \rightarrow 1} \frac{1 + \cos(\pi x)}{\tan^2(\pi x)}$$

$$(h) \lim_{h \rightarrow 0} \frac{\sin(x + h) - \sin x}{h}$$

$$(i) \lim_{h \rightarrow 0} \frac{\cos(x + h) - \cos x}{h}$$

$$(j) \lim_{x \rightarrow 0} \frac{\arcsin x}{x}$$

$$(k) \lim_{x \rightarrow 1} \frac{\sin(\pi x)}{\sin(3\pi x)}$$

Exercise 1.6. Let $f(x) = \frac{1}{x-1} - \frac{1}{1-x^3}$.

(1) Find the domain of f .

(2) Can you compute $\lim_{x \rightarrow 1} f(x)$? Why?

- (3) Describe the function $g(x)$ given by $g(x) = f(1+x)$.
- (4) Compute $\lim_{x \rightarrow 0^+} g(x)$ and $\lim_{x \rightarrow 0^-} g(x)$.
- (5) Use the previous item to compute $\lim_{x \rightarrow 1^+} f(x)$ and $\lim_{x \rightarrow 1^-} f(x)$.

Exercise 1.7. Find all the pairs of real numbers a and b that satisfy simultaneously

$$\lim_{x \rightarrow 0} \frac{\sqrt{ax^2 + bx + 1} - 1}{x} = 3 \quad \text{y} \quad \lim_{x \rightarrow +\infty} \frac{\sqrt{ax^2 + bx + 1} - 1}{x} = 2.$$

Exercise 1.8. For each of the following cases compute $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$. Determine if $\lim_{x \rightarrow 0} f(x)$ exists and make the graph of the function.

(a) $f(x) = |3x - 6|$

(b) $f(x) = \frac{|x|}{x}$

(c) $f(x) = \begin{cases} -x + 5 & x \leq 0 \\ x^2 + 1 & x > 0 \end{cases}$

Exercise 1.9. Compute the following limits

(a) $\lim_{x \rightarrow 3} (3x - 5)^{\frac{1}{1-x}}$

(d) $\lim_{x \rightarrow 0} \left(\frac{\sin(2x)}{x} \right)^{\frac{\tan x}{3x}}$

(b) $\lim_{x \rightarrow \infty} \left(\frac{3x+1}{2x-5} \right)^{\frac{x+1}{3x+1}}$

(e) $\lim_{x \rightarrow +\infty} \left(\frac{\sqrt{2x^2 + 1} + 1}{x} \right)^{x+1}$

(c) $\lim_{x \rightarrow 0^+} \left(\frac{\sin(2x)}{\sin x} \right)^{\frac{1}{x}}$

(f) $\lim_{x \rightarrow 0^+} \left(\frac{\sin(3x^2)}{\sin(4x^2)} \right)^{\frac{1}{x}}$

Exercise 1.10. Using that $\lim_{y \rightarrow 0} (1+y)^{\frac{1}{y}} = \lim_{t \rightarrow \infty} \left(1 + \frac{1}{t}\right)^t = e$, compute the following limits

(a) $\lim_{x \rightarrow +\infty} \left(\frac{x-2}{x+3} \right)^x$

(e) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$

(b) $\lim_{x \rightarrow +\infty} \left(1 + \frac{a}{x}\right)^x, a \in \mathbb{R}$

(f) $\lim_{h \rightarrow 0} \frac{\ln(a+h) - \ln a}{h}, a > 0$

(c) $\lim_{h \rightarrow 0} \left(1 + \frac{h}{x}\right)^{\frac{1}{h}}$

(g) $\lim_{h \rightarrow 0} \frac{e^h - 1}{h}$

(d) $\lim_{x \rightarrow 0^+} (1 + \sin x)^{\frac{1}{x}}$

(h) $\lim_{h \rightarrow 0} \frac{e^{a+h} - e^a}{h}, a \in \mathbb{R}$

Exercise 1.11. Compute the following limits

$$(a) \lim_{x \rightarrow \pi} \frac{1 - \sin(\frac{x}{2})}{\pi - x}$$

$$(d) \lim_{x \rightarrow 0} \frac{x - \sin^2 x}{x + \sin 2x}$$

$$(b) \lim_{x \rightarrow +\infty} x(\sqrt{x+a} - \sqrt{x-a})$$

$$(e) \lim_{x \rightarrow +\infty} x \sin \frac{1}{x}$$

$$(c) \lim_{x \rightarrow 0} \frac{1}{x} \ln \sqrt{\frac{1+x}{1-x}}$$

$$(f) \lim_{x \rightarrow +\infty} \frac{1}{\sqrt{x^2 + x} - x}$$

Exercise 1.12. Let $a, b \in \mathbb{R}$. Show that $\lim_{t \rightarrow \infty} (1 + \frac{a}{t} + \frac{b}{t^2})^t$ does not depend on the choice of b .

Exercise 1.13. Study the one sided limits and the continuity at the indicated values for x . Describe which types of discontinuities you find.

$$(a) f(x) = \frac{\operatorname{tg} x}{x} \quad x = 0$$

$$(c) f(x) = \frac{1}{1 + e^{\frac{1}{x}}} \quad x = 0$$

$$(b) f(x) = \begin{cases} \frac{\sin x}{x} & x \neq 0 \\ 0 & x = 0 \end{cases} \quad x = 0$$

$$(d) f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0 \end{cases} \quad x = 0$$

Exercise 1.14. Determine for which values of A we get a continuous function

$$f(x) = \begin{cases} \frac{x^2-4}{x-2} & \text{if } x \neq 2 \\ A & \text{if } x = 2 \end{cases}$$

Exercise 1.15. Determine all the points of discontinuity of the following functions, and redefine them to get continuous functions, if possible.

$$(a) f(x) = \frac{x-1}{x(x^2-4)}$$

$$(b) f(x) = \begin{cases} x & x < 0 \\ x^2 & 0 \leq x < 2 \\ 2 & x \geq 2 \end{cases}$$

$$(c) f(x) = \frac{x^{\frac{2}{3}} - 4}{2x^{\frac{2}{3}} - 3x^{\frac{1}{3}} - 2}$$

$$(d) f(x) = \frac{(x-1)^2}{x^2-1}$$

$$(e) f(x) = \begin{cases} 4x^2 - 3 & x > 1 \\ 1 & x = 1 \\ \frac{x^2 - 3x + 2}{x^2 - 4x + 3} & x < 1 \end{cases}$$

$$(f) f(x) = \frac{x^2}{\sqrt{x^2+1}-1}$$

$$(g) f(x) = \begin{cases} \frac{\sqrt{3x+1} - \sqrt{x+3}}{x^2-x} & x > 1 \\ \frac{\operatorname{sen}(-2x+2)}{x^2+x-2} & x < 1 \end{cases}$$

Exercise 1.16. Determine all the pairs of numbers a and b such that the following functions are continuous

$$(a) f(x) = \begin{cases} x & x \in (-\infty, 0] \\ ax + b & x \in (0, 2) \\ x^2 & x \geq 2 \end{cases}$$

$$(b) f(x) = \begin{cases} x^3 + 1 & x \leq 0 \\ ax^2 + b & 0 < x < 2 \\ x^2 - 1 & x \geq 2 \end{cases}$$

$$(c) f(x) = \begin{cases} \frac{\sqrt{x+3} - \sqrt{3x+1}}{\sqrt{x-1}} & x > 1 \\ ax + b & -2 \leq x \leq 1 \\ \frac{x^2 + 2x}{x^2 + x - 2} & x < -2 \end{cases}$$

Exercise 1.17. Determine all the points of discontinuity of the following functions, and redefine them to get continuous functions, if possible.

$$(a) f(x) = \begin{cases} \frac{e^x}{x-1} & x > 0 \\ 0 & x = 0 \\ \frac{x^2+2x-3}{x+3} & x < 0 \end{cases} \quad (c) f(x) = \begin{cases} \frac{x\sqrt{x+2}}{\sqrt{4x+1}-3} & x > 2 \\ 1 & x = 2 \\ \frac{9x-18}{x^2+4x-12} & x < 2 \end{cases}$$

$$(b) f(x) = \begin{cases} \frac{\sqrt{x+9}-3}{\sqrt{x}} & x > 0 \\ 2 & x = 0 \\ \frac{1-\cos x}{x^2+x} & x < 0 \end{cases}$$

Exercise 1.18. Study the continuity at $x = 5$ of the function

$$f(x) = \begin{cases} 3xe^{x-5} & \text{if } x \leq 5, \\ \frac{45 \sin(x-5) \ln(\frac{4}{5}x-3)}{2(x-5)} & \text{if } x > 5. \end{cases}$$

Study the continuity of this function on $\mathbb{R} \setminus \{5\}$.

Exercise 1.19. Show that the polynomial $P(x) = x^3 + x + 1$ is such that $P(a) = 0$ for some $a \in (-1, 0)$.

Exercise 1.20. Show that the intersection of the graphs of the functions $f(x) = e^x$ and $g(x) = x + 2$ for $x \geq 0$ is non empty.

Exercise 1.21. Show that there exists some $a \in (1, e)$ such that $\frac{\ln a}{a} = \frac{1}{3}$.

Exercise 1.22. Determine the existence of values a in $[-4, -3]$, $[-3, 3]$ and $[-1, 1]$ such that $f(a) = 0$ for $f(x) = \frac{|x|}{4-x^2}$.

2. DERIVATIVES

Exercise 2.1. *In each of the following cases, compute by definition, if possible, the derivative of the function at the indicated points.*

(a) $f(x) = x^2 + 1, \quad x = 1$

(b) $g(x) = x^3 + 2, \quad x = 1$

(c) $h(x) = \sqrt{2x-1}, \quad x = 5$

Exercise 2.2.

(1) Let $f(x) = \frac{1}{x^2}$. Find $f'(a)$ for any $a \neq 0$.

(2) Let $g(x) = e^{-x}$. Compute $g(0) + g'(0)$.

(3) Let $h(x) = 3 - x$ and $m(x) = 1 - \sin \frac{\pi x}{2}$. Compute $\frac{m'(1)}{h'(1)}$.

Exercise 2.3. *Compute, by definition, the derivative of the following functions.*

(a) $f(x) = x^2 + 1$

(f) $f(x) = \sqrt{x}$

(b) $f(x) = x^3$

(g) $f(x) = \sin x$

(c) $f(x) = 2x^3 + 3x^2 + x - 2$

(h) $f(x) = \cos x$

(d) $f(x) = \frac{3}{x+4}$

(i) $f(x) = e^x$

(e) $f(x) = \frac{2}{x^2}$

(j) $f(x) = \ln x$

Exercise 2.4.

(1) Let $f(x) = |x| + x$. Compute, if they exist, $f'(1)$, $f'(-2)$ and $f'(0)$.

(2) Let $g(x) = x|x|$. Compute, if they exist, $g'(1)$, $g'(-2)$ and $g'(0)$.

(3) Find $g'(x)$.

Exercise 2.5. *Let $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ given by*

$$f(x) = |x|, \quad g(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}, \quad h(x) = \begin{cases} x^2 & \text{if } x \leq 0 \\ 0 & \text{if } x > 0 \end{cases}$$

(1) Show that the functions are continuous at $x = 0$.

(2) Make the graph of the functions.

(3) Prove that f and g are not differentiable at $x = 0$.

(4) Study the existence of $h'(0)$.

Exercise 2.6. Compute $f'(x)$ using rules of derivation.

$$(a) f(x) = x^5 - 4x^3 + 2x - 3$$

$$(j) f(x) = \frac{2 + \cos x}{3 + \sin x}$$

$$(b) f(x) = (x + 2)(x + 3)(3x + 1)(2 + 5x^2)$$

$$(k) f(x) = \frac{\sqrt{x} + x}{x^2 + 1}$$

$$(c) f(x) = 7\cos x + 5\sin x + xe^x$$

$$(l) f(x) = \frac{\sin x + \cos x}{x \ln x}$$

$$(d) f(x) = 2x \sin x - (x^2 - 2)e^x$$

$$(m) f(x) = \frac{1}{x^2 + 1} - \sin x$$

$$(e) f(x) = e^x \cos x + \ln 3$$

$$(f) f(x) = x^2 \ln x - \frac{x^3}{3} + e^x$$

$$(n) f(x) = \frac{1}{x} + 2 \ln x - \frac{\ln x}{x}$$

$$(g) f(x) = x(3 + x^2) + \ln 2$$

$$(\tilde{n}) f(x) = \frac{x}{1 + \sqrt{x}} + \frac{\sqrt{x}}{1 + x}$$

$$(h) f(x) = \frac{1}{x} + \frac{1}{\sqrt{x}} + \frac{1}{\sqrt[3]{x}}$$

$$(o) f(x) = \frac{x + e^x}{5 - x} + \operatorname{tg} x$$

$$(i) f(x) = \frac{x - 1}{x}$$

$$(p) f(x) = (\ln x \log_a x) - (\ln a \log_a x)$$

Exercise 2.7. Compute $f'(x)$ using the chain rule.

$$(a) f(x) = (1 + x)^{129}$$

$$(h) f(x) = e^{x^2 + 3x}$$

$$(b) f(x) = \cos(3x)$$

$$(i) f(x) = \ln(e^x + \sin 5x)$$

$$(c) f(x) = \ln(\sin x)$$

$$(j) f(x) = 2^{\sin x}$$

$$(d) f(x) = \operatorname{tg}(5x^5)$$

$$(k) f(x) = 3^{\cos x} + \sin^2 x$$

$$(e) f(x) = \ln(x^5)$$

$$(l) f(x) = e^{\sin x} - 3\pi e^{\operatorname{tg} x}$$

$$(f) f(x) = 3 \sin^4 x$$

$$(m) f(x) = (a + bx^4)^{\frac{1}{3}}$$

$$(g) f(x) = \ln^5 x$$

Exercise 2.8. .

(1) Let $f(x) = x^3 + x$. Compute $(f^{-1})'(0)$ and $(f^{-1})'(2)$.

(2) Let $f(x) = x^9 + x^5 + 2x^3 + 3x + 5$. Compute $(f^{-1})'(5)$.

Exercise 2.9. Compute the derivative of the following functions.

$$(a) f(x) = \arcsin x$$

$$(d) f(x) = \operatorname{arcc} \operatorname{tg} x$$

$$(b) f(x) = \arccos x$$

$$(e) f(x) = \arcsin \sqrt{x}$$

$$(c) f(x) = \arctan x$$

$$(f) f(x) = \ln(\arccos x)$$

Exercise 2.10. Using the logarithm, compute the derivative of the following functions.

(a) $f(x) = x^{3x}$

(f) $f(x) = \sqrt[3]{x}$

(b) $f(x) = (\sqrt{x+1})^x$

(g) $f(x) = (\sin x)^{x^2}$

(c) $f(x) = (\ln x)^x$

(h) $f(x) = (\ln x)^{x^2+x}$

(d) $f(x) = (\ln x)^{\ln x}$

(i) $f(x) = (\sin^3 x)^{\ln x}$

(e) $f(x) = (\cos x)^{e^x}$

(j) $f(x) = x^{(x^x)}$

Exercise 2.11. Compute the derivative of the following functions.

(a) $f(x) = x^{\sin x} + \sqrt{\sin 3x}$

(b) $f(x) = x \left((\sin x)^{x^2} + \pi e^x \right)$

(c) $f(x) = \frac{x e^{(\sqrt[3]{x} + \cos x - 1)}}{\sin x + \ln(x^2 + e)} + (\cos x)^{\frac{1}{x}} + (\sqrt{2})^\pi$

Exercise 2.12. Consider the functions $f(x) = x^3 + 2$ and $g(x) = 2x^2 + 2$.

(1) Prove that the graphs of these functions intersect in two points.

(2) Study the behaviour of the tangent lines for both curves at these points.

Exercise 2.13. Determine in which points of the curve for $y = \ln x$, the tangent line is parallel to the line that goes through $(1, 0)$ and $(e, 1)$.

Exercise 2.14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x^3$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ another function with $g'(2) = 4$. Compute $(g \circ f)'(1)$.

Exercise 2.15. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(g(x^2 + x)) + 3g(x) = 3 \tan(x) + 1$, $g(0) = 0$ and $g'(0) = 2$. Compute $f'(0)$ and find the tangent line of f at $x = 0$.

Exercise 2.16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ the function given by

$$f(x) = \begin{cases} \sqrt{|x-1|} & \text{if } x \geq 2 \\ \cos(x-2) & \text{if } x < 2 \end{cases}$$

Determine if f is continuous at $x = 2$ and if it is differentiable at $x = 1$ and at $x = 2$.

Exercise 2.17. Determine if the following functions are continuous and differentiable.

$$(a) f(x) = \begin{cases} 1-x & x \leq 0 \\ e^{-x} & x > 0 \end{cases}$$

$$(e) f(x) = \begin{cases} (x+1)^2 & x \leq -1 \\ \frac{\cos(x\pi)}{\pi} & x > -1 \end{cases}$$

$$(b) f(x) = \begin{cases} x^2 & x \leq 3 \\ 3x & x > 3 \end{cases}$$

$$(f) f(x) = \begin{cases} \frac{3-x^2}{2} & x \leq 1 \\ \frac{1}{x} & x > 1 \end{cases}$$

$$(c) f(x) = \begin{cases} x^2 & x \leq 3 \\ 2x+3 & x > 3 \end{cases}$$

$$(g) f(x) = \begin{cases} x^2 \operatorname{sen} \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$(d) f(x) = |(x-1) \cdot (x+1)|$$

$$(h) f(x) = \sqrt{|x|} \operatorname{sen}(4|x|)$$

Exercise 2.18. Find all $a, b, c \in \mathbb{R}$ such that $f'(1)$ exists, for

$$f(x) = \begin{cases} \ln x & \text{if } x \geq 1 \\ a(x-1)^2 + b(x-1) + c & \text{if } x < 1 \end{cases}$$

Exercise 2.19. Compute the second and the third derivative of the following functions.

$$(a) f(x) = 3x^3 + 5x - 1$$

$$(d) f(x) = (x^2 + 1)^5$$

$$(b) f(x) = \ln(7x)$$

$$(e) f(x) = \operatorname{sen}(4x)$$

$$(c) f(x) = e^{-x}$$

$$(f) f(x) = \cos(2x^3)$$

Exercise 2.20. .

(1) Compute the derivatives of order 7 and 8 of $f(x) = x^7 - 5x^4 + 8x$.

(2) Compute the derivatives of order n and $n+1$ of a polynomial of degree n .

(3) Compute the derivatives of order n of $f(x) = \sin x$ and $g(x) = \cos x$, for any natural number n .

3. L'HOSPITAL RULE - EXTREMES

Exercise 3.1. Compute the following limits, using L'Hospital rule, if it is possible.

$$(a) \lim_{x \rightarrow 1} \frac{\ln x}{1-x}$$

$$(b) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

$$(c) \lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{3x^2 + 2x - 5}$$

$$(d) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$$

$$(e) \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x - \sin x}$$

$$(f) \lim_{x \rightarrow 0} \frac{e^{2x} - e^{-2x} - 4x}{x - \sin x}$$

$$(g) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln(\sin x)}{\pi - 2x}$$

$$(h) \lim_{x \rightarrow 0^+} \frac{\ln(\sin x)}{\pi - 2x}$$

$$(i) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan x}{\tan 5x}$$

$$(j) \lim_{x \rightarrow +\infty} \frac{e^x}{x}$$

$$(k) \lim_{x \rightarrow -\infty} \frac{e^x}{4x^5}$$

$$(l) \lim_{x \rightarrow 0^+} x^2 e^{\frac{1}{x}}$$

$$(m) \lim_{x \rightarrow 0^-} x^2 e^{\frac{1}{x}}$$

$$(n) \lim_{x \rightarrow +\infty} \frac{\ln x}{x^4}$$

$$(\tilde{n}) \lim_{x \rightarrow 0^+} x \ln \frac{1}{x}$$

$$(o) \lim_{x \rightarrow +\infty} \frac{\ln x}{e^x}$$

$$(p) \lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{\cos x}{\sin x} \right)$$

$$(q) \lim_{x \rightarrow 1^+} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$$

$$(r) \lim_{x \rightarrow 0} \frac{\ln x}{\cotg x}$$

$$(s) \lim_{x \rightarrow 0} (x - \sin x) \ln x$$

$$(t) \lim_{x \rightarrow +\infty} \frac{x + \ln x}{x \ln x}$$

$$(u) \lim_{x \rightarrow 0} \left(\cotg x - \frac{1}{x} \right)$$

$$(v) \lim_{x \rightarrow 0^+} x^x$$

$$(w) \lim_{x \rightarrow 0} (1+x)^{\cotg x}$$

$$(x) \lim_{x \rightarrow 0^+} x^{\sin x}$$

$$(y) \lim_{x \rightarrow +\infty} x^{\frac{1}{x}}$$

$$(z) \lim_{x \rightarrow +\infty} (\ln x)^{\frac{1}{1-\ln x}}$$

Exercise 3.2. Let $f : [-2, 2] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2 & \text{if } x \in [-2, 1] \\ 3x - 2 & \text{if } x \in (1, 2]. \end{cases}$$

Prove that f is continuous, that $f(-2) = f(2)$, that $f'(0) = 0$ and that $f'(1)$ does not exist.

Is this a contradiction to Rolle's Theorem?

Exercise 3.3. Let $f(x) = x(x+1)(x+2)(x+3)$. Show that the equation $f'(x) = 0$ has exactly three solutions in $\mathbb{R}$.

Exercise 3.4. Using Lagrange's Theorem, prove the following inequalities.

- (a) $\operatorname{sen} x \leq x, \quad \forall x \geq 0.$
- (b) $|\operatorname{sen} x - \operatorname{sen} y| \leq |x - y|, \quad \forall x, y \in \mathbb{R}.$
- (c) $\ln(1+x) < x, \quad \forall x > 0.$

Exercise 3.5. *Is it possible to apply L'Hospital rule in order to compute the following limit?*

$$\lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{\sin x}$$

If your answer is YES, compute it. If your answer is NO, explain why and compute the limit using another method.

Exercise 3.6. *Determine if the following functions are differentiable.*

$$(a) f(x) = \begin{cases} 3xe^{x-4} & x \leq 4 \\ \frac{20 \operatorname{sen}(x-4) \ln(\frac{3}{4}x-2)}{x-4} + 12 & x > 4 \end{cases}$$

$$(b) f(x) = \begin{cases} x \left(\frac{1}{3x^2+2x} \right)^{\operatorname{sen} x} & x > 0 \\ x & x \leq 0 \end{cases}$$

Exercise 3.7. *Compute $f'(0)$ if it exists:*

$$(a) f(x) = \begin{cases} 1 & x \leq 0 \\ (\cos x)^{\frac{1}{x}} & x > 0 \end{cases} \quad (b) f(x) = \begin{cases} 1+x^2 & x \leq 0 \\ x^{\operatorname{sen}^2 x} & x > 0 \end{cases}$$

Exercise 3.8. *Determine the global extremes of the following functions:*

- (1) $f(x) = 3x + 1$ in $[-1, 3]$;
- (2) $f(x) = \sin(2x)$ in $[0, \pi]$;
- (3) $f(x) = |x - 2|$ in $[-1, 1]$.

Exercise 3.9. *Determine if x_0 is a local extreme in each of the following cases:*

- (a) $f(x) = x^2 + 1, \quad x_0 = 0.$
- (b) $f(x) = |x - 2|, \quad x_0 = 2.$
- (c) $f(x) = x^6 + 5, \quad x_0 = 0.$
- (d) $f(x) = x^3, \quad x_0 = 0.$

Exercise 3.10. *Determine the local extremes of the following functions:*

(a) $f(x) = x^3 - 6x + 2$

(e) $f(x) = x + \frac{1}{x} \quad (x \neq 0)$

(b) $f(x) = x^2 e^{-x}$

(f) $f(x) = \frac{x+1}{(x-1)^2} \quad (x \neq 1)$

(c) $f(x) = x \ln x \quad (x > 0)$

(g) $f(x) = \frac{2x}{1+x^2}$

(d) $f(x) = x^{\frac{2}{3}}(1-x)$

Exercise 3.11. Study the domain of the following functions; determine their local extremes, increasing and decreasing intervals, points of inflection and make the corresponding graphs.

(a) $f(x) = 12x^2(x+1)$

(k) $f(x) = \frac{x+2}{\ln(x+2)}$

(b) $f(x) = \frac{1}{4}x^4 - \frac{3}{2}x^2$

(l) $f(x) = e^x(x^2+2)$

(c) $f(x) = \frac{1-x^3}{x}$

(m) $f(x) = \frac{2x^3}{x^2-4}$

(d) $f(x) = x^{\frac{2}{3}}(1-x) \quad [-1; 1]$

(n) $f(x) = \frac{x^2-4}{x^2-3}$

(e) $f(x) = \frac{x^2-1}{x}$

(\tilde{n}) $f(x) = \frac{x^2-4}{x^3}$

(f) $f(x) = \frac{x^2+2x-3}{x^2}$

(o) $f(x) = \frac{x^2}{\sqrt{x+1}}$

(g) $f(x) = \frac{1}{x^2+1}$

(p) $f(x) = \frac{x(x-3)}{(x+3)^2}$

(h) $f(x) = e^{-x^2}$

(q) $f(x) = xe^x$

(i) $f(x) = x \ln x$

(r) $f(x) = x^2 + |x|$

(j) $f(x) = xe^{-x}$

Exercise 3.12. Find the extremes (local and global) of the following functions. Make the graphs.

$$(a) \quad f(x) = \begin{cases} 2x^2 - 2x - 2 & \text{if } x \leq 2 \\ x & \text{if } x > 2 \end{cases}, \quad (b) \quad f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ (x-2)^2 & \text{if } x > 1 \end{cases}$$

Exercise 3.13. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that is differentiable in its domain and satisfies the following conditions:

(1) $f'(-1) = f'(-\frac{1}{2}) = f'(0) = f'(\frac{3}{2}) = 0;$

(2) $\{x \in \mathbb{R} : f'(x) > 0\} = (-\infty, -1) \cup (0, \frac{3}{2});$

(3) $\{x \in \mathbb{R} : f'(x) < 0\} = (-1, -\frac{1}{2}) \cup (-\frac{1}{2}, 0) \cup (\frac{3}{2}, +\infty).$

Determine the local extremes for this function. Justify. Make a graph of a function satisfying these conditions.

Exercise 3.14. Prove the following inequalities

(a) $\sin x \geq x \quad \forall x \leq 0.$

(d) $1 - \cos x \leq \frac{x^2}{2} \quad \forall x \in \mathbb{R}.$

(b) $e^x \geq 1 + x \quad \forall x \in \mathbb{R}.$

(c) $\ln x \leq 2(\sqrt{x} - 1) \quad \forall x > 1.$

(e) $\ln(1+x) > \frac{x}{1+x} \quad \forall x > 0.$

Exercise 3.15. .

(1) Show that the function $f(x) = ax^3 + bx^2 + cx + d$, with $a \neq 0$, can have at most two local extremes.

(2) Give an example of a function like this with two local extremes.

(3) Give an example of a function like this without extremes.

(4) Can you find an example of a function like this with exactly one local extreme?
Why?

4. INTEGRATION

Exercise 4.1. Find a function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

- | | |
|-------------------------------|-------------------------|
| (i) $g'(x) = 2$ | (v) $g'(x) = x^2$ |
| (ii) $g'(x) = x$ | (vi) $g'(x) = e^x$ |
| (iii) $g'(x) = \text{sen } x$ | (vii) $g'(x) = x + x^2$ |
| (iv) $g'(x) = \cos x$ | (viii) $g'(x) = x^n$ |

Are they unique? In each item, find g such that $g(0) = 5$.

Exercise 4.2. Verify the $F(x) + C$ are primitives for $f(x)$ in each of the following cases:

- | | |
|---|--|
| (a) $f(x) = x^\alpha, \quad \alpha \neq -1$ | $F(x) = \frac{x^{\alpha+1}}{\alpha+1}$ |
| (b) $f(x) = \frac{1}{x}$ | $F(x) = \ln x $ |
| (c) $f(x) = e^x$ | $F(x) = e^x$ |
| (d) $f(x) = \text{sen } x$ | $F(x) = -\cos x$ |
| (e) $f(x) = \cos x$ | $F(x) = \text{sen } x$ |
| (f) $f(x) = \frac{1}{\sqrt{1-x^2}}$ | $F(x) = \text{arc sen } x$ |
| (g) $f(x) = \frac{-1}{\sqrt{1-x^2}}$ | $F(x) = \text{arc cos } x$ |
| (h) $f(x) = \frac{1}{1+x^2}$ | $F(x) = \text{arc tg } x$ |
| (i) $f(x) = \frac{1}{\cos^2 x}$ | $F(x) = \text{tg } x$ |

Exercise 4.3. Compute the following primitives

- | | |
|--|---|
| (a) $\int x^2 dx$ | (e) $\int \left(\frac{1}{x} + 2x - e^x \right) dx$ |
| (b) $\int x^{100} dx$ | (f) $\int x^{-\frac{1}{2}}(3x + \sqrt{x}) dx$ |
| (c) $\int \sqrt{x} dx$ | (g) $\int \left(x^2 + \frac{1}{x} \right)^2 dx$ |
| (d) $\int (3x - \cos x + 2 \text{ sen } x) dx$ | (h) $\int \frac{1}{\cos^2 x \text{ sen}^2 x} dx$ |

Exercise 4.4. Compute the following primitives using substitutions.

- | | |
|---|---|
| (a) $\int 5 \cos(5x) dx$ | (j) $\int x(x^2 + 1)^{-1} dx$ |
| (b) $\int \cos(x + 5) dx$ | (k) $\int \cos x \operatorname{sen}^{-2} x dx$ |
| (c) $\int \operatorname{sen}(7x) dx$ | (l) $\int (5 - 2x)^{-1} dx$ |
| (d) $\int (t + 1)(t^2 + 2t + 3)^{\frac{2}{3}} dt$ | (m) $\int \cos^{-2}(2x) dx$ |
| (e) $\int e^{3x} dx$ | (n) $\int x^{-1} \cos(\ln x) dx$ |
| (f) $\int (x + 1)^{-1} dx$ | (ñ) $\int x(1 + 3x^2)^{-1} dx$ |
| (g) $\int x^{-1} \ln x dx$ | (o) $\int (1 + y^{\frac{1}{2}})^{\frac{1}{2}} dy$ |
| (h) $\int (x + 1)^{-1} \ln(x + 1) dx$ | (p) $\int (x^2 + 2x + 5)^{-1} dx$ |
| (i) $\int x e^{x^2} dx$ | (q) $\int x(16 + x^4)^{-1} dx$ |

Exercise 4.5. Compute the following primitives using integration by parts.

- | | |
|--------------------------------------|---|
| (a) $\int x \operatorname{sen} x dx$ | (d) $\int x \ln x dx$ |
| (b) $\int (x^2 - 2)e^{-x} dx$ | (e) $\int x^2(x + 4)^{-\frac{1}{2}} dx$ |
| (c) $\int x^3 \cos x dx$ | (f) $\int (t^2 + t)(t + 1)^{-5} dt$ |

Exercise 4.6. Compute the following primitives using simple rational fractions.

- | | |
|--|---|
| (a) $f(x) = \frac{3x - 2}{(x + 2)(x + 3)}$ | (f) $f(x) = (x^2 - 1)^{-2}$ |
| (b) $f(x) = 4(x - 2)^{-1}(x - 1)^{-1}$ | (g) $f(x) = \frac{7x - 6}{3x^3 + 6x^2 + 3x}$ |
| (c) $f(x) = \frac{2x + 2}{x^2 - 4}$ | (h) $f(x) = \frac{8x^3 + 7}{8(x + 1)(x + \frac{1}{2})^2}$ |
| (d) $f(x) = \frac{x^3 - 2x}{x^2 - 1}$ | (i) $f(x) = \frac{2(2x^2 + 1)}{(x^2 + 1)(x - 1)}$ |
| (e) $f(x) = \frac{x}{(x + 1)(x - 2)(x + 3)}$ | (j) $f(x) = \frac{3}{(x^2 + 2x + 3)(x + 2)}$ |

Exercise 4.7. Compute the following primitives

- | | |
|--|---|
| (a) $\int \frac{3x^2}{\sqrt{5x^3+1}} dx$ | (k) $\int x^5 \sqrt[5]{5-x^2} dx$ |
| (b) $\int x^{\frac{1}{2}} \ln x dx$ | (l) $\int \operatorname{sen}^5\left(\frac{x}{6}\right) \cos\left(\frac{x}{6}\right) dx$ |
| (c) $\int \frac{\ln^3 x}{x} dx$ | (m) $\int \frac{x - \sqrt{\operatorname{arctg} 2x}}{1+4x^2} dx$ |
| (d) $\int x \operatorname{arc} \operatorname{sen} x dx$ | (n) $\int \ln(x^2+1) dx$ |
| (e) $\int (1+\cos 2x)^{-2} \operatorname{sen} x dx$ | (ñ) $\int \frac{e^x + 3e^{2x}}{1-e^{2x}} dx$ |
| (f) $\int \frac{(\operatorname{sen} x - 2)\cos x}{\operatorname{sen}^2 x + \operatorname{sen} x - 2} dx$ | (o) $\int e^{\sqrt{x}} dx$ |
| (g) $\int \frac{xe^{3x^2}}{4+e^{3x^2}} dx$ | (p) $\int \frac{\ln x + 1}{x(\ln^2 x - 3 \ln x)} dx$ |
| (h) $\int \operatorname{tg} x \ln(\cos x) dx$ | (q) $\int (x^3+x)\cos(x^2+1) dx$ |
| (i) $\int x(\operatorname{sen} x + \frac{x}{2}\cos x)\sqrt{x^2\operatorname{sen} x} dx$ | (r) $\int \sqrt{e^x+1} dx$ |
| (j) $\int \frac{x}{\sqrt{1+x^2+(1+x^2)^{\frac{3}{2}}}} dx$ | (s) $\int \frac{\cos^3 x}{1+\cos^2 x} dx$ |
| | (t) $\int \operatorname{tg}^3 x dx$ |

Exercise 4.8.

- (1) Find $f(x)$ such that $f'(x) = 2(\sin x)^3 \cos x$ and $f(0) = 3$.
- (2) Find $g(x)$ such that $g'(x) = x \sin(5x)$ and $g(\frac{\pi}{2}) = 1$.
- (3) Find $h(x)$ such that $h'(x) = \frac{x+6}{x^2-4}$ and $h(3) = 0$.

Exercise 4.9. Find a differentiable function $f(x)$ such that f has a critical point at $x = 2$, the point $(2, 3)$ belongs to the graph of f and $f'(x) = -x + a$ for some $a \in \mathbb{R}$.

Exercise 4.10. Using Barrow's rule, compute

- | | | |
|-------------------------------|--------------------------|--|
| (a) $\int_0^1 e^t dt$ | (c) $\int_0^x \cos t dt$ | (e) $\int_{\frac{\pi}{2}}^x \operatorname{sen} t dt$ |
| (b) $\int_1^2 \frac{1}{t} dt$ | (d) $\int_1^x t^r dt$ | |

Exercise 4.11. Applying substitution and integration by parts, compute the following definite integrals

$$(a) \int_{-1}^{\sqrt{2}} x(x^2 + 1)^{-3} dx$$

$$(e) \int_1^4 x^{-1} \cos(\ln x) dx$$

$$(b) \int_0^{\frac{\pi}{4}} (1 + \cos 2x)^{-2} \sin(2x) dx$$

$$(f) \int_0^1 e^x (4 + 3e^x) dx$$

$$(c) \int_3^7 x \ln x dx$$

$$(g) \int_3^9 x \sqrt{x+1} dx$$

$$(d) \int_0^2 (1+x^2)(x-1)^{\frac{2}{3}} dx$$

$$(h) \int_0^1 e^x (e^{2x} + e^x + 1)^{-1} dx$$

Exercise 4.12. Compute the derivative of following functions

$$(a) f_1(x) = \int_1^x e^{-t^2} dt$$

$$(e) f_5(x) = \int_0^x \frac{\sin t}{1+t} dt$$

$$(b) f_2(x) = \int_0^x \sqrt{1+t^2} dt$$

$$(f) f_6(x) = \int_0^{\sin x} \frac{t}{2+t^2} dt$$

$$(c) f_3(x) = \int_0^{x^2} \frac{\cos t}{1+t^2} dt$$

$$(g) f_7(x) = \int_x^{x^3} \cos(t^2) dt$$

$$(d) f_4(x) = \int_2^{\operatorname{tg} x} \operatorname{arc} \operatorname{tg} t dt \quad x \in \left(-\frac{\pi}{2}; \frac{\pi}{2}\right)$$

$$(h) f_8(x) = \int_{\cos x}^{e^x} t^2 dt$$

Exercise 4.13. Find the local extremes of the function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$g(x) = \int_0^x e^{t^2} t^2 (t-1)(t-4) dt$$

Exercise 4.14. Compute the area of the region limited by

$$(1) y = \frac{1}{2}x^2, y = 0, x = 1 \text{ and } x = 2.$$

$$(2) y = -x + 5, y = 0, x = 6 \text{ and } x = 9.$$

Exercise 4.15. Compute the area of the region limited by

$$(1) y = x^3, y = 0, x = 2 \text{ and } x = 5.$$

$$(2) y = x^2 - x + 1, y = 0, x = 0 \text{ and } x = 1.$$

Exercise 4.16. Determine $a \in \mathbb{R}_{>0}$ such that the area of the region limited by $y = \sin x$, $y = 0$, $x = 0$ and $x = a$ is $\frac{5}{2}$.

Exercise 4.17. Determine the area of the region limited by the following curves

$$(a) \ y = 3x^2 - 7x, \ y = x^2 + x - 6$$

$$(b) \ y^2 = x, \ x - y = 2$$

$$(c) \ y = x^{1/3}, \ x = 0, \ y = 1$$

$$(d) \ y = e^x, \ y = e^{-x}, \ x = 1, \ x = -1$$

$$(e) \ y = x^{1/2}, \ y = x - 2, \ x = 0$$

$$(f) \ y = x^2 - 1, \ y = -x$$

$$(g) \ y = x^3 - x, \ y = x$$

5. SERIES

Exercise 5.1. *Prove that*

- (1) $\sum_{n=1}^{\infty} \frac{5}{n(n+1)} = 5$
- (2) $\sum_{n=1}^{\infty} 2 = +\infty$
- (3) $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \frac{1}{2}$
- (4) $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}+\sqrt{n-1}} = +\infty$
- (5) $\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4}$
- (6) $\sum_{n=1}^{\infty} \frac{1}{(4n-1)(4n+3)} = \frac{1}{12}$
- (7) $\sum_{n=1}^{\infty} \frac{1}{n+1} = +\infty$
- (8) $\sum_{n=1}^{\infty} \frac{2^n+3^n}{6^n} = \frac{3}{2}$
- (9) $\sum_{n=1}^{\infty} 2\left(\frac{6}{5}\right)^n = +\infty$

Exercise 5.2. *Prove that the real number $0.444\dots$ is a rational number.*

Exercise 5.3. *Prove that $2.999\dots = 2 + \sum_{n=1}^{\infty} \frac{9}{10^n} = 3$.*

Exercise 5.4. *Prove that the real number $2.020202\dots$ is a rational number.*

Exercise 5.5. *Determine if the following series are convergent or not:*

- (1) $\sum_{n=1}^{\infty} \frac{n^2}{2n^2+1}$
- (2) $\sum_{n=1}^{\infty} \frac{n+1}{2n+1}$
- (3) $\sum_{n=1}^{\infty} \frac{n+1}{n^3}$
- (4) $\sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}}$
- (5) $\sum_{n=1}^{\infty} \frac{(n+3)n}{3n^2+n-2}$
- (6) $\sum_{n=1}^{\infty} \frac{1}{n2^n}$
- (7) $\sum_{n=1}^{\infty} \frac{2^n}{n}$
- (8) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$
- (9) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2n+1}}$
- (10) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(2n+1)}}$
- (11) $\sum_{n=1}^{\infty} \frac{n+\sqrt{n}}{2n^3-1}$
- (12) $\sum_{n=1}^{\infty} \frac{2^n}{3^n+n}$

Exercise 5.6. *Determine if the following series are convergent or not:*

- (1) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$
- (2) $\sum_{n=1}^{\infty} \frac{2n-1}{(\sqrt{2})^n}$

$$(3) \sum_{n=1}^{\infty} \frac{n!}{(n+1)^n}$$

$$(4) \sum_{n=1}^{\infty} \frac{2^n}{n(n+1)}$$

$$(5) \sum_{n=1}^{\infty} \frac{1}{n^n}$$

$$(6) \sum_{n=1}^{\infty} \left(\frac{n}{n^2+2}\right)^n$$

$$(7) \sum_{n=1}^{\infty} \left(\frac{n+1}{2n-1}\right)^n$$

$$(8) \sum_{n=1}^{\infty} \frac{2^n}{n^n}$$